THE CATHOLIC DIOCESE OF TANGA
TANGA CATHOLIC SECONDARY SCHOOLS UNION (TACASSU)
FORM FOUR JOINT PRE-NATIONAL EXAMINATION 1

033/2A BIOLOGY 2A
(ACTUAL PRACTICAL 2A)
3HRS ADVANCED INSTRUCTIONS 2026

IMPORTANT

wazaelimu.com

Great care must be taken not to divulge these instructions to both candidates and unauthorized person either directly or indirectly.

Make sure that the candidates are provided with **SPECIMENS, CHEMICALS AND APPARATII** indicated in this instructions only and not otherwise

QUESTION 1

Specimens and Apparatus
Provide each candidate with:

Apparatus

1. Four clean test tubes labelled:
- O
- 1
- 2
- 3

1. One thermometer
2. One test tube holder
3. One beaker containing warm water (35–37°C)
4. One dropper
5. One measuring cylinder or syringe
6. One source of heat:
- spirit burner/Bunsen burner/hot water bath

1. One rubber band

Chemicals/Reagents

Solution Q

Provide:

- 10 cm^3 starch solution
- Label as "Q"

Benedict's Solution

- About 10 cm^3 per candidate

Iodine Solution

- About 5 cm^3 per candidate

Distilled or tap water

- Sufficient quantity

QUESTION 2

Specimens

Provide the following labelled specimens:

E = Tilapia

F = Earthworm

G = Frog

H = Pine leaf/twig

Specimens may be:

- fresh,
- preserved,
- mounted,
- or clear photographs/models if unavailable.

Additional Laboratory Requirements

Provide:

- Tissue paper
- Waste container
- Handwashing water
- Soap/disinfectant

Preparation Notes for Teachers/Laboratory Technicians

1. Prepare fresh starch solution before the examination.
2. Ensure Benedict's solution is uncontaminated.
3. Ensure water bath temperature remains between 35–37°C.
4. Confirm all specimens are clearly labelled.
5. Ensure enough boiling water/heating facilities are available.
6. Test all reagents before the practical session.